

BRIAN TANG

🔗 <https://www.bjaytang.com>

AI Security & Privacy Researcher

Mission: Seeking a role as a Research Scientist or Product-Minded Researcher in AI Security, Safety, and Privacy. I build attacks, defenses, and products for LLMs, VLMs, and coding agents, and love turning research into real-world product impact. My expertise lies at the intersection of AI, human factors, product, and security & privacy.

EDUCATION

Ph.D. Candidate | *Computer Science and Engineering*

University of Michigan - Ann Arbor

Thesis: Security and Privacy Challenges with Vision-Language Models and Smart Glasses.

Fall 2021 – Present

Advised by [Kang G. Shin](#)

Bachelor of Science | *Major: Computer Science*

University of Wisconsin - Madison

Fall 2017 – Winter 2020

Advised by [Kassem Fawaz](#), [Varun Chandrasekaran](#), [Somesh Jha](#)

EXPERIENCE

Research Intern

Summer 2026

IBM Research - Thomas J. Watson Research Center

Manager: [Ian Molloy](#), Mentor: [Fred Araujo](#)

- Securing intent drift, prompt injection, secret egress, memory poisoning for sandboxed coding agents. Using reference monitors, plan-time sketches, and effect system graphs to create task-level dynamic permission scoping.
- Ran comprehensive long-horizon modified evals (e.g., SWE-Bench Pro, Terminal-Bench 2.0) on sandboxed Claude Code.

Graduate Research Assistant

Fall 2021 – Present

University of Michigan

Committee: [Kang G. Shin](#), [Florian Schaub](#), [Ke Sun](#), [Stella Yu](#), [Alex Halderman](#)

- Creating automated data collection and annotation methods for training vision-language models to achieve above-human performance on in-the-wild text recognition tasks (>10k samples). [10]
- Designed and patented a real-time software privacy screen, Eye-Shield. Reduced attack success rates from 100% to 24.24% for images and from 77.27% to 15.91% for text, protecting against screen snooping on smartphones. [3]
- Developed various LLM web automation tools and document parsers for auditing data collection activities [12]. Deployed automated analysis of 47.2k Chrome Web Store extensions [4], 2.9k online trackers [7], and 1.4k cookie banners [1], finding many instances of misleading disclosures and non-compliance.
- Built and evaluated an LLM advertising chatbot using GPT-4o and GPT-3.5. Our user study with 179 participants found that ads influenced 13.07% more participants, with 19.05% more having positive reactions to products. Discovered that serving ads decreases LLM performance on math, reasoning, and reading comprehension tasks by 2-3%. [2]
- Served as server admin, social organizer, equipment curator, and mentoring role for the Real-Time Computing Lab.

Undergraduate Research Assistant

Fall 2018 – Spring 2021

University of Wisconsin - Madison

Advisors: [Kassem Fawaz](#), [Somesh Jha](#), Mentor: [Varun Chandrasekaran](#)

- Researched fairness properties of face recognition systems. [5]
- Created a controller for social robots to preserve conversational privacy. [6]
- Explored using physical invariants from LiDAR to improve ML classifier robustness against adversarial attacks. [15]
- Developed an anti face recognition system using adversarial attacks to protect online photo privacy. [8]

Software Engineering Intern

Summer 2019

Roblox Corporation

Manager: Tyler Berg

- Designed and implemented production autocomplete, smart-cursor movement, and suggestion features for Roblox Studio's script editor using TDD. Integrated 30 JavaScript Squish tests to auto-validate UI behavior and prevent errors.

Software Engineering Intern

Summer 2018

Optum, UnitedHealth Group

Manager: Kyle Erickson

- Designed and implemented an attack-surface visualization that aggregated and normalized 50M+ vulnerability and asset records, produced correlated risk scores, and an interactive dashboard. Presented results to Optum's leadership.

PUBLICATIONS

- [1] Brian Tang, Duc Bui, and Kang G. Shin. “Navigating Cookie Consent Violations Across the Globe”. **34th USENIX Security Symposium**. *Acc Rate: 17%*. 2025. [\[paper\]](#).
- [2] Brian Tang, Kaiwen Sun, Noah T. Curran, Florian Schaub, and Kang G. Shin. “Ads that Talk Back: Implications and Perceptions of Injecting Personalized Advertising into LLM Chatbots”. **Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (UbiComp/IMWUT)**. *Acc Rate: 20%*. 2025. [\[paper\]](#).
- [3] Brian Tang and Kang G. Shin. “Eye-Shield: Real-Time Protection of Mobile Device Screen Information from Shoulder Surfing”. **32nd USENIX Security Symposium**. *Acc Rate: 17%*. 2023. [\[paper\]](#).
- [4] Duc Bui, Brian Tang, and Kang G. Shin. “Detection of Inconsistencies in Privacy Practices of Browser Extensions”. **44th IEEE Symposium on Security and Privacy**. *Acc Rate: 13%*. 2023. [\[paper\]](#).
- [5] Harrison Rosenberg, Brian Tang, Kassem Fawaz, and Somesh Jha. “Fairness Properties of Face Recognition and Obfuscation Systems”. **32nd USENIX Security Symposium**. *Acc Rate: 17%*. 2023. [\[paper\]](#).
- [6] Brian Tang, Dakota Sullivan, Bengisu Cagiltay, Varun Chandrasekaran, Kassem Fawaz, and Bilge Mutlu. “Confidant: A Privacy Controller for Social Robots”. **17th ACM/IEEE International Conference on Human-Robot Interaction**. *Acc Rate: 26%*. 2022. [\[paper\]](#).
- [7] Duc Bui, Brian Tang, and Kang G. Shin. “Do Opt-Outs Really Opt Me Out”. **29th ACM Conference on Computer and Communications Security**. *Acc Rate: 18%*. 2022. [\[paper\]](#).
- [8] Varun Chandrasekaran, Chuhan Gao, Brian Tang, Kassem Fawaz, Somesh Jha, and Suman Banerjee. “Face-Off: Adversarial Face Obfuscation”. **21st Privacy Enhancing Technologies Symposium**. *Acc Rate: 22%*. 2021. [\[paper\]](#).

PATENTS

Real-Time Protection For Mobile Devices From Shoulder Surfing [3]	Spring 2023
U.S. Pat. App. No. 63/468,650-Conf. #8672	Filed

GRANTS

An Efficient Real-Time Knowledge Base for Smart Glasses and Smartphones Samsung (START); Converted to joint proposal w/ Ke Sun , Anhong Guo , Kang G. Shin Ideated Project with Liangkai Liu . Wrote 1/2 of the Proposal.	Spring 2025 Granted, \$200k, 3-Pages
I-SEE: Intelligent Vehicular Perception and Control General Motors R&D Ideated Project with Postdoc and 2 Ph.D. Students. Wrote 1/2 of the Proposal.	Spring 2025 Granted, \$145k, 3-Pages
Evaluating Privacy and Surveillance Risks of Large Language Models National Artificial Intelligence Research Resource Pilot (NAIRR, \$10k) Outlined and Scoped Projects. Wrote Full Proposal.	Winter 2025 Granted, \$10k, 3-Pages
Securing Interactions between Driver and Vehicle Using Batteries National Science Foundation (NSF) Cloud Credits (Cloudbank) Wrote 2 Pages and Midterm Reports.	Summer 2023 Granted, \$16k, 2-Pages
Securing Cyber-Physical System Communication and Control Defense University Research Instrumentation Program (DURIP). Initiated and Organized Proposal Structure and 25 Equipment Orders. Wrote 1/3 of the Proposal.	Spring 2023 Granted, \$300k, 19-Pages

PREPRINTS

- [9] [Brian Tang, Fred Araujo, Xiaokui Shu, and Ian Molloy. “GRAPNEL: Securing Intent Drift in Coding Agents”. **In Preparation: 36th USENIX Security Symposium** \(2027\).](#)
- [10] [Brian Tang, Qingyu Zhu, and Kang G. Shin. “Hawkeye: Reading Illegible Text with Vision Language Models”. **In Preparation: Annual Meeting of the Association for Computational Linguistics \(ACL\)** \(2026\).](#)
- [11] [Noah T. Curran, Brian Tang, and Kang G. Shin. “Motion-Based Visual Instruction Tuning”. **In Submission: IEEE/CVF Conference on Computer Vision and Pattern Recognition \(CVPR\)** \(2026\).](#)
- [12] [Brian Tang and Kang G. Shin. “Steward: Natural Language Web Automation”. **arXiv** \(2024\). \[\\[paper\\]\]\(#\).](#)
- [13] [Noah T. Curran, Minkyung Cho, Ryan Feng, Liangkai Liu, Brian Tang, Pedram Mohajer Ansari, Alkim Domeke, Mert D. Pesé, and Kang G. Shin. “Short: Achieving the Safety and Security of the End-to-End AV Pipeline”. **1st Cyber Security in Cars Workshop \(CSCS\) at CCS**. *Acc Rate: 39%*. 2024. \[\\[paper\\]\]\(#\).](#)
- [14] [Bulut Gozubuyuk, Brian Tang, Mert D. Pesé, and Kang G. Shin. “I Know What You Did \(In Your Car\) Last Summer: Privacy Implications of Android Automotive OS”. **arXiv** \(2024\). \[\\[paper\\]\]\(#\).](#)
- [15] [Varun Chandrasekaran, Brian Tang, Nicolas Papernot, Kassem Fawaz, Somesh Jha, and Xi Wu. “Rearchitecting Classification Frameworks For Increased Robustness”. **arXiv** \(2020\). \[\\[paper\\]\]\(#\).](#)

HONORS AND AWARDS

Bloomberg Summer of Puzzles Competition (<i>Finalist</i>) Puzzle Hunt Competition	Spring 2024 Free Trip to NYC HQ
Travel Grants Rackham, UMich CoE, USENIX	2023 – 2025 \$4k
3 Minute Thesis Competition (<i>Finalist</i>) Recovering Privacy and Autonomy in the Era of Large Language Models	Fall 2023 1 of 26 Finalists
College of Engineering Fellowship University of Michigan 1st year PhD fellowship	Fall 2021 \$90k
CVS Health Foundation Program Scholarship (Outstanding Children of CVS Employees)	Fall 2017 \$5k

PRESENTATIONS

Implications and Perceptions of Injecting Personalized Advertising into LLM Chatbots [2] FTC, Washington D.C. Third Federal Trade Commission Conference on Marketing and Public Policy	Spring 2026 Event Link
CS47500 Human-Computer Interaction (Guest Lecturer) Purdue University Taught HCI Students About AI Data Collection Studies, AI Security & Privacy User Studies (1 Hour)	Spring 2026
Navigating Cookie Consent Violations Across The Globe [1] Seattle, WA <i>USENIX Security Symposium</i>	Aug 2025
Defending Against Deepfakes and Disinformation (Guest Lecturer) University of Michigan Law School Taught 30 Law Students About ML, GANs, Deepfakes, Stable Diffusion (1.5 Hours)	Fall 2024 Link to Slides
Steward: Natural Language Web Automation [12] Ann Arbor, MI <i>SECRT Security Reading Group</i>	Mar 2024
Recovering Privacy and Autonomy in the Presence of Language Models Ann Arbor, MI <i>3 Minute Thesis Finalist Competition (Engineering Graduate Symposium)</i>	Sept 2023

Eye-Shield: Real-Time Protection of Mobile Device Screen Information from Shoulder Surfing [3]	Aug 2023
Anaheim, CA <i>USENIX Security Symposium</i>	
Confidant: A Privacy Controller for Social Robots [6]	Mar 2022
The Internet <i>ACM/IEEE International Conference on Human-Robot Interaction</i>	
Face-Off: Adversarial Face Obfuscation [8]	Jan 2021
The Internet <i>VMWare - NSF: Data Privacy and Edge Computing</i>	
Face-Off: Adversarial Face Obfuscation [8]	July 2021
The Internet <i>Proceedings on Privacy Enhancing Technologies Symposium</i>	

INTERESTS

Artificial Intelligence: Adversarial ML, Computer Vision, NLP, DNNs, CNNs, (M)LLMs, Agents, VLMs, RAG
Security and Privacy: Usable Privacy, Online Privacy, Face Recognition, Social Privacy, Mobile Privacy, Surveillance
Human-Computer Interaction: User Studies, Social Robotics, Mobile Computing, Real-Time & Cyber-Physical Systems

SKILLS

Programming: Python, JavaScript, HTML, SQL, GLSL, C++ , Kotlin, TeX TeX, Linux, Bash, Godot
Software Development: GitHub, Perforce, Qt, NginX, Flask, Squish, AWS, Redis, PostgreSQL, OpenGL, d3.js, Electron
Machine Learning: TensorFlow, PyTorch, Keras, Pandas, NumPy, HuggingFace, Transformers, YOLO, Llama, PEFT
Languages: English (Native), Chinese Mandarin (Spoken-Only), Japanese (Beginner), French (Beginner)
Flight Experience: Cessna 172 – 2hrs Cessna 152 – 2hrs
Hobbies & Interests: Reading, Hiking, Meditation, Camping, Music Production, Videogames, Tabletop RPGs

SERVICE

External/Sub Reviewer	Spring 2020 – Summer 2026
USENIX Security, PoPETS, NeurIPS, CHI, Ubicomp, HAI, IEEE Privacy	
	9 Papers
Poster PC Committee Member	Spring 2024 – Spring 2025
IEEE S&P 2024-2025	
Co-Chair/Organizer	Fall 2025
Prof. Kang G. Shin’s Retirement Symposium	
87 Attendees, 3 Distinguished Speakers (Atul Prakash , Mingyan Liu , Farnam Jahanian)	